

1 Problem Statement

Students today are faced with an overwhelming amount of learning material, such as lecture slides, PDFs, and notes. Traditional study methods like re-reading and highlighting are largely passive and have been shown to be inefficient for long-term knowledge retention. While techniques such as active recall and spaced repetition are proven to significantly improve learning outcomes, they are often underutilized due to the manual effort required to create and manage study materials like flashcards.

Additionally, students frequently encounter difficulties in understanding complex concepts and must rely on external resources, which disrupts their study flow. Existing tools often fail to provide an integrated solution that combines efficient memorization techniques with personalized explanations.

This project aims to address these challenges by developing an AI-powered study assistant that automates the creation of flashcards from lecture materials, optimizes review scheduling using spaced repetition algorithms, and provides on-demand explanations to support deeper understanding.

2 Main Functionality

The application provides the following core functionalities:

- Automatic generation of flashcards from uploaded lecture materials (e.g., PDFs, text).
- Implementation of spaced repetition (SuperMemo-2 algorithm) to optimize review intervals.
- Interactive study sessions based on active recall principles.
- AI-powered explanations that simplify complex concepts on demand.

3 Target Users

The primary target users of the application are:

- University students preparing for exams.
- Students in knowledge-intensive fields such as STEM, medicine, and law.

Secondary users may include lifelong learners and participants in online courses who seek efficient study methods.

4 GenAI Integration

Generative AI is integrated into the system in two key ways:

1. **Flashcard Generation:** The system processes raw lecture materials and converts them into structured flashcards in a question-answer format. This significantly reduces the manual effort required for content preparation.
2. **Concept Explanation:** During study sessions, users can request simplified explanations of flashcards. The AI generates clear, concise explanations, potentially including examples or analogies, enabling better understanding without leaving the application.

5 Usage Scenarios

5.1 Exam Preparation

A student uploads lecture material into the system. The application automatically generates flashcards and schedules them for review using a spaced repetition algorithm. The student engages in study sessions where they actively recall answers and rate difficulty, allowing the system to adapt future review intervals.

5.2 Understanding Difficult Concepts

While studying, a student encounters a flashcard they do not fully understand. The student requests an explanation, and the AI provides a simplified and more intuitive version of the concept, improving comprehension.

5.3 Active Study Sessions

The system presents flashcards that are due for review. The student attempts to recall the answer, reveals it, and rates their performance. The system updates the review schedule accordingly, ensuring efficient long-term retention.

5.4 Flexible AI Deployment

The system supports both cloud-based and local AI models. Users can choose between high-performance cloud models or privacy-preserving local models, depending on their needs.